



**Republic of the Philippines**  
**Department of Education**  
National Capital Region  
Schools Division Office of Taguig City and Pateros  
**TAGUIG SCIENCE HIGH SCHOOL**

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# **EYEDRIVESAFE: A PORTABLE IoT-ENABLED DRIVER DROWSINESS DETECTION AND MONITORING SYSTEM USING TRANSFER LEARNING-BASED FACIAL BEHAVIORAL ANALYSIS**

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**Presented to the Faculty of Senior High School**

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## ABSTRACT

### **EyeDriveSafe: A PORTABLE IoT-ENABLED DRIVER DROWSINESS DETECTION AND MONITORING SYSTEM USING TRANSFER LEARNING-BASED FACIAL BEHAVIORAL ANALYSIS**

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Drowsy driving is a critical road safety issue in the Philippines, where road traffic deaths rose 39% from 2011 to 2021 (WHO, 2023). This study presents EyeDriveSafe, a portable, non-intrusive drowsiness detection system deploying a MobileNetV2 0.35 convolutional neural network through Edge Impulse's transfer learning platform on an Arduino Uno Q microcontroller. Trained on 8,930 images, the model achieved a validation accuracy of 97.7%, validation loss of 0.07, 70% controlled simulation accuracy, and AUC-ROC of 0.98. The system uses whole-frame binary classification to derive the Drowsy Frame Ratio (DFR) and Drowsy Episode Count (DEC), feeding a three-tier alert state machine (AWAKE-WARNING-ALERT) with time-delay false alarm prevention. Additional features include a real-time Flask web dashboard, Arduino IoT Cloud remote monitoring via MQTT, and an AI-powered fatigue report via GPT-4.1-mini. Performance evaluations confirmed strong validation metrics and alert reliability, with a critical limitation under dark conditions where drowsy-state detection failed. Survey results among 30 respondents yielded a product performance rating of 4.54 out of 5, indicating strong consumer acceptance. EyeDriveSafe demonstrates that advanced driver safety technology can be deployed affordably in the Philippine transportation context, supporting SDG 3 and SDG 11.

Keywords: *driver drowsiness detection, transfer learning, MobileNetV2, IoT, Arduino, Drowsy Frame Ratio*

## **I. INTRODUCTION**

### **Rationale and Background of the Study**

Road traffic accidents remain a perennial social issue affecting public safety and national development in the Philippines. Despite sustained government interventions, road-related injuries and fatalities continue to rise, posing serious consequences for communities and the economy. According to the World Health Organization (2023), road traffic deaths in the Philippines increased by 39% from 2011 to 2021, with young adults comprising a significant proportion of victims. This persistent public safety issue underscores the urgent need for innovative and preventive road safety solutions.

Within this broader concern, driver fatigue and drowsiness have emerged as a specific and critical problem. Local and regional studies consistently show that Filipino drivers, particularly those in public transportation, are exposed to prolonged driving hours, irregular schedules, and inadequate rest. Such working conditions significantly increase the risk of drowsiness-related accidents, which often go undetected until it is too late. Recent transportation safety reviews emphasize that drowsy driving is comparable to alcohol impairment in its effect on reaction time, attention, and decision-making. Yet, it remains under-monitored in developing countries such as the Philippines (Al-Quraishi, 2024).

To address this problem, global solutions for driver drowsiness detection have been developed. These include physiological sensor-based systems (e.g., EEG and ECG), wearable devices, and rule-based camera monitoring systems. While physiological approaches provide detailed fatigue indicators, recent studies note that they are intrusive, uncomfortable, and impractical for daily or long-haul driving (Sahayadhas et al., 2021). Wearable devices also suffer from user compliance issues and high cost. Meanwhile, traditional computer vision

systems relying on handcrafted features often struggle with varying lighting conditions, head movements, and facial diversity, limiting their real-world reliability (Albadawi et al., 2022).

In response to these limitations, this study proposes EyeDriveSafe, a portable, non-intrusive driver drowsiness detection system that performs edge AI inference directly on the Arduino Uno Q, representing a clear research gap in the Philippine context. Unlike conventional approaches that require server-grade hardware or cloud connectivity for deep learning inference, EyeDriveSafe deploys a MobileNetV2 convolutional neural network through Edge Impulse's transfer learning platform to classify facial drowsiness indicators in real time using a lightweight camera setup. Deep learning is emphasized in this study due to its proven ability to automatically learn complex visual features, maintain high accuracy under variable lighting and motion conditions, and generalize better across diverse facial profiles, key requirements for real-world driving environments (Essahraoui et al., 2025; Hassan et al., 2025).

Recent literature supports the effectiveness of deep learning-based vision systems for drowsiness detection. Fonseca et al. (2025) demonstrated that CNN-based eye-state classification significantly outperforms traditional threshold-based methods in real-world driving scenarios. Similarly, Hassan et al. (2025) found that transformer-based eye-closure detection models achieve high accuracy while remaining computationally efficient for embedded, in-vehicle systems. Albadawi et al. (2022) further highlight that camera-based behavioural monitoring, when combined with modern deep learning techniques, offers a balanced solution between accuracy, affordability, and user comfort.

The relevance of EyeDriveSafe extends beyond technological innovation and directly contributes to the United Nations Sustainable Development Goals (SDGs), particularly SDG 3: Good Health and Well-Being, which aims to reduce global deaths and injuries from road traffic accidents, and SDG 11: Sustainable Cities and Communities, which emphasizes safe, accessible, and sustainable transport systems. By providing an affordable and context-appropriate drowsiness detection solution, EyeDriveSafe supports safer mobility and promotes public well-being, especially for Filipino drivers and commuters.

By integrating recent advances in deep neural networks into a portable and accessible driver-monitoring system, this study addresses a critical gap between high-end research solutions and the practical road safety needs of the Philippine transportation sector.

### **Research Objectives:**

This research aims to develop a driver drowsiness detection system designed to detect and alert drowsy drivers to prevent accidents caused by fatigue through deep learning and IoT integration. With this, the following objectives are determined:

1. To develop and deploy a machine learning–based drowsiness detection pipeline capable of classifying driver fatigue states in real time.
  - a. Training and optimizing an image classification model to distinguish between "awake" and "drowsy" states using whole-frame facial analysis on the Arduino Uno Q microcontroller.
  - b. Computing drowsiness indicators, specifically the Drowsy Frame Ratio (DFR) and drowsy episode count, derived from continuous model inference to quantify fatigue levels over time.

- c. Implementing a three-tier state machine (AWAKE, WARNING, and ALERT) that translates computed drowsiness indicators into escalating system responses, including buzzer and LED-based alerts.
- 2. To assess the performance of the EyeDriveSafe system through the following tests:
  - a. A drowsy state accuracy test to evaluate the system's ability to correctly classify awake and drowsy states under controlled conditions.
  - b. A validation metrics evaluation using Edge Impulse to assess the model's precision, recall, F1 score, and AUC-ROC performance.
  - c. A confusion matrix analysis to determine the model's per-class classification accuracy in correctly identifying awake and drowsy states
  - d. An on-device performance evaluation to measure the inference time, peak RAM usage, and Flash storage requirements of the deployed model on the Arduino Uno Q.
- 3. To design and implement an IoT-integrated monitoring and reporting system that extends the detection capabilities beyond the device itself.
  - a. Integrating Arduino IoT Cloud connectivity to enable real-time remote monitoring of driver fatigue states and session data through a web-based dashboard.
  - b. Developing a web dashboard with real-time streaming capabilities for live visualization of detection outputs, drowsiness frame ratio values, and system states.
  - c. Implementing an LLM-powered fatigue analysis module that generates intelligent session reports based on accumulated drowsiness data, delivered through the Arduino IoT Cloud Messenger widget.

- d. An alert response time test to measure the latency between drowsiness detection and the activation of the buzzer and LED alert mechanisms.
- 4. To conduct a cost analysis for the production, distribution, and projected retail pricing of the proposed device.
- 5. To evaluate the marketability and potential consumer acceptance of the product among its target users.

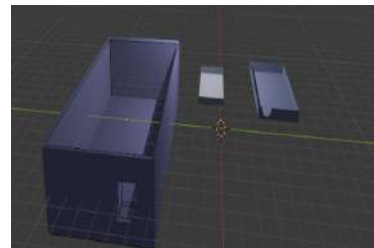
## II. MATERIALS AND PROCEDURE

The procedure below was designed and implemented to achieve the provided objectives. Materials were also described using the provided procedure. The following procedure was designed to achieve objectives, prepare materials, construct devices, and connect electrical components.

### Materials

The electrical components required for the EyeDriveSafe system, such as the Arduino Uno Q and the Camera Module for eye-closure monitoring, will be purchased from online stores operating within Manila, Philippines. In contrast, the materials needed for the device's external housing and mounting mechanism, including the chassis, will be acquired from physical hardware stores located within Taguig City, Metro Manila, Philippines. The researchers will also utilize tools and equipment they already possess for assembly, which include the soldering iron, resistor, wires, cables, and screws.

A. **3D Printed Box.** A custom-fabricated enclosure produced through additive manufacturing serves as the protective housing responsible for securing and organizing the internal electronic components of the EyeDriveSafe system.



B. **Arduino Uno Q.** A hybrid single-board computer (SBC) integrating a Qualcomm QRB2210 microprocessor (MPU) running Debian Linux and an STM32U585 microcontroller (MCU) for real-time control, serving as the main processing unit for the EyeDriveSafe drowsiness detection system and its alert mechanism.





C. **Cables and Connectors.** A hardware interface device designed to extend a USB-C connection, serving as the data transmission link responsible for connecting the web camera to the processing unit of the EyeDriveSafe system.



D. **Modulino Buzzer.** A modular piezoelectric buzzer from the Arduino Modulino family, connected via QWIIC interface, serving as the auditory warning component responsible for alerting the driver when drowsiness is detected



E. **Modulino Pixels.** A modular LED strip from the Arduino Modulino family, connected via QWIIC interface, serving as the visual warning component responsible for providing color-coded status indication (green for awake, yellow for warning, red for alert) during system operation.



F. **Type C USB Hub.** Insulated wires and plugs are used to transfer power and data between components. They form the electrical pathways linking all the subsystems together.



G. **Web Camera.** The primary input sensor of the system. This device captures the video feed or still images of the driver's face. It provides the raw visual data needed for the drowsiness detection algorithm.



## Tools and Software

A. **Arduino App Lab.** A specialized software application designed for configuring Arduino boards, serving as the setup and configuration tool responsible for initializing and managing the Arduino Uno Q within the EyeDriveSafe system.



B. **Arduino IoT Cloud.** A cloud-based Internet of Things management platform used for remote device monitoring and data visualization, serving as the remote monitoring infrastructure responsible for transmitting real-time drowsiness metrics to authorized users via MQTT protocol.



C. **Arduino IDE.** An integrated development environment designed for writing, compiling, and uploading code to Arduino boards, serving as the programming platform responsible for implementing and controlling the functional logic of the EyeDriveSafe system.



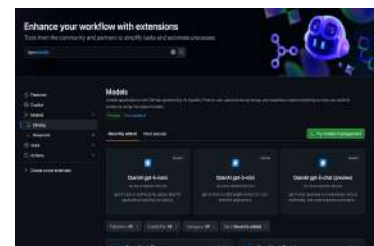
D. **Edge Impulse.** An artificial intelligence development platform with pre-trained models and deployment tools, serving as the deep learning framework responsible for training, optimizing, and integrating neural network-based drowsiness detection algorithms into the EyeDriveSafe system.



**E. Flask.** A lightweight Python web framework, serving as the local web server responsible for hosting the real-time monitoring dashboard accessible on the same network as the device.



**F. GitHub Models API.** A free-tier cloud-hosted large language model inference service providing access to GPT-4.1-mini, serving as the artificial intelligence engine responsible for generating post-session fatigue analysis reports from logged driving data.



**G. Laptop and Computer.** A general-purpose computing device used for software development and data processing, serving as the primary platform responsible for programming the Arduino Uno Q and training the deep neural network models used in the EyeDriveSafe system.



**H. Python 3.13.** A high-level programming language used for machine learning and system integration, serving as the main programming language responsible for implementing, testing, and deploying the deep learning models for driver drowsiness detection.



**I. Socket.IO** is a real-time, bidirectional event-based communication library, serving as the data streaming framework responsible for delivering live detection metrics from the processing unit to the local web monitoring dashboard

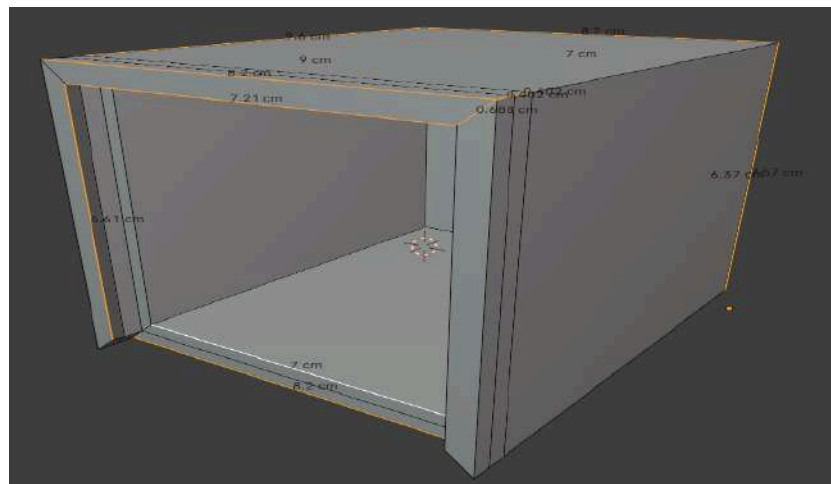


## Procedures

### I. Designing EyeDriveSafe

The initial stage focused on the conceptual and physical design of the EyeDriveSafe system. A three-dimensional product enclosure was designed using Blender to produce a housing suitable for 3D printing. The enclosure was designed to securely accommodate the Arduino Uno Q, USB camera, Modulino Buzzer, and Modulino LED in a compact and portable form.

*Figure 1. 3D model of the chassis of EyeDriveSafe*



### II. Acquisition of Materials

Following the design phase, all required electronic and mechanical components were acquired. These included the Arduino Uno Q, USB camera, Modulino Buzzer, Modulino LED, and 3D Printed Box. All materials were selected based on compatibility, functionality, and suitability for embedded machine learning applications. Each component was inspected and tested before assembly to ensure proper operation and system reliability.

|                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                  |                                                                                                |                                                                                                                  |                                                                                                   |                                                                                                  |
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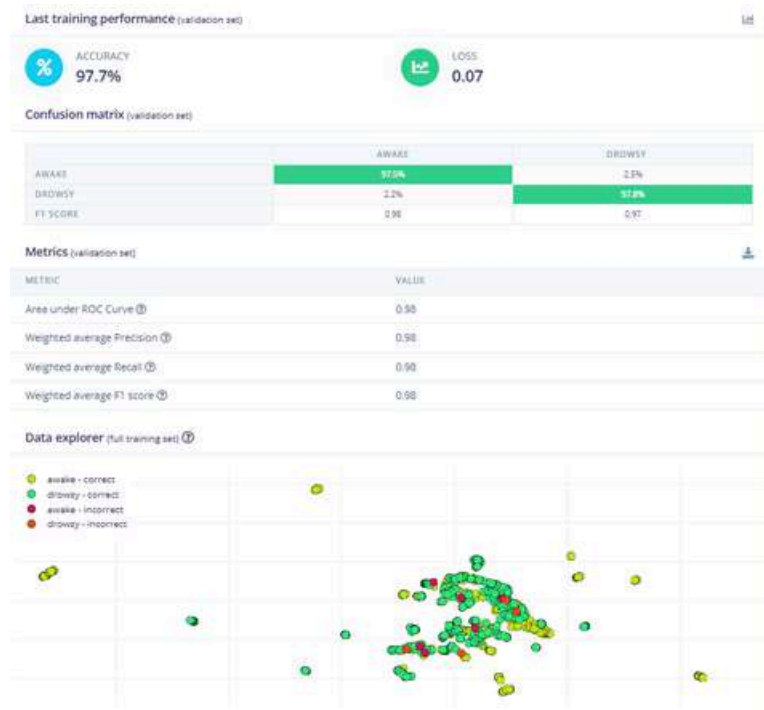
Model preparation was conducted using the Edge Impulse platform. A project workspace was established using the Transfer Learning block with MobileNetV2 0.35 as the backbone architecture. Image preprocessing was configured with 96×96 greyscale input, reducing the feature vector from 230,400 (color) to 76,800 (greyscale), a deliberate design choice to reduce model size and improve robustness to color-temperature variations in different lighting environments.

Data collection was conducted in two phases. In Phase 1, a convenience sampling method was used to physically collect 1,782 images from researchers, relatives, and friends



effective learning and reliable classification performance. The iterative development process is documented in Table 1 (Model Iteration History).

*Figure 5. MobileNetV2 0.35 Model Last Training and Validation Performance*



## VI. Mathematical Modeling of Drowsiness Metrics

The foundational metric for drowsiness detection, PERCLOS, measures the continuous percentage of eyelid closure in a clinical or high-fidelity setting. Because the EyeDriveSafe system processes discrete video frames and outputs binary classification (open/closed) to optimize computational efficiency on portable hardware, a mathematical translation from a continuous time domain to a discrete sequential space is mandatory. This derivation preserves the established scientific validity of PERCLOS while defining the operational parameters for the system's specific fatigue indicators: the Drowsy Frame Ratio (DFR) and Drowsy Episode Count (DEC).

### 1. Foundational Metric: PERCLOS

The standard PERCLOS computation over a video sequence is given by:

$$P = \frac{E_c}{E_t}$$

Where  $E_c$  is the count of frames classified as eyes-closed and  $E_t$  is the total frame count within the observation interval.

## 2. Derivation of Drowsy Frame Ratio (DFR)

To adapt PERCLOS for the binary, whole-face classification approach of the EyeDriveSafe system, the 80% pupil coverage criterion is replaced with a discrete frame-level drowsiness classification output.

Let  $d(i)$  represent the discrete frame-level classification for frame  $i$ :

$$d(i) = \begin{cases} 1, & \text{if frame is classified as drowsy} \\ 0, & \text{if frame is classified as awake} \end{cases}$$

A frame is classified as drowsy when the model's drowsy-class confidence score exceeds a user-configurable threshold  $T$ , with a default value of  $T = 0.50$ . Formally,  $d(i) = 1$  if  $P(\text{drowsy}|\text{frame}_i) \geq T$ , and  $d(i) = 0$  otherwise. For a rolling temporal window containing  $N$  total frames, the DFR translates the continuous PERCLOS formula into a discrete mathematical ratio:

$$\text{DFR} = \frac{1}{N} \sum_{i=1}^N d(i)$$

## 3. Derivation of Drowsy Episode Count (DEC)

To provide a secondary fatigue indicator, the system tracks discrete drowsy episodes. An episode is isolated when a classification transition violates a specific duration threshold, operating independently of the rolling ratio.



Let  $d(t)$  represent the drowsiness state classification at time  $t$ . An episode is detected, and the DEC increments when the following transition sequence occurs:

$$\begin{aligned} d(t_{\text{onset}}) & \text{ transitions from } 0 \rightarrow 1 \\ d(t_{\text{offset}}) & \text{ transitions from } 1 \rightarrow 0 \end{aligned}$$

Subject to the maximum duration constraint  $B$ :

$$t_{\text{offset}} - t_{\text{onset}} \leq B$$

Where  $B = 0.6$  seconds.

This value is derived from the physiological duration of a normal eye blink, which ranges from 0.1 to 0.4 seconds (Schleicher et al., 2008). Setting  $B$  at 0.6 seconds provides a margin above the upper blink duration while excluding sustained drowsiness events, which are instead handled by the state machine alert system. Transitions exceeding  $B$  indicate prolonged drowsiness rather than momentary lapses. Episode frequency over the rolling window is computed as:

$$EF(t) = \frac{E_w}{W} \times 60$$

Where  $E_w$  is the count of qualifying episodes within window  $W$  (in seconds), normalized to episodes per minute.

## VII. Software Development

A three-tier progressive alert state machine was designed to translate model output into escalating system responses. The AWAKE state resets after 0.5s of awake classification. After 1.5s of sustained drowsy frames, the system transitions to WARNING (yellow LED). After 3.0s total, the system escalates to ALERT (red LED + Modulino Buzzer). This

time-delay design prevents false alarms from single-frame misclassifications. The Python backend computes Drowsy Frame Ratio (DFR), the ratio of drowsy-classified frames to total frames over a 60-second rolling window, and drowsy episode count, defined as the number of distinct continuous drowsy sequences. These replace the previously cited PERCLOS and blink rate metrics, which require facial landmark detection that the binary classification model does not perform. A local web dashboard (Flask + Socket.IO) provides real-time visualization. Arduino IoT Cloud integration enables remote monitoring of Alert State, Drowsy Episodes, Alert Count, and Fatigue Report variables via MQTT. An LLM-powered post-session fatigue report is generated using GPT-4.1-mini via the GitHub Models API.

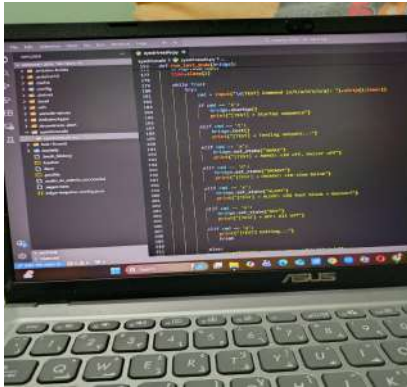
*Figure 6. Arduino IoT Cloud Interface of EyeDriveSafe*



## VIII. System Integration and Assembly

Following model selection, the trained neural network was exported as an .eim file and deployed to the Arduino Uno Q. System integration involved programming the detection process using Python for model inference and image classification, while C++ (Arduino) was used to manage hardware functionalities such as system initialization and activation of the alert mechanism. The system was configured to automatically start upon being powered on, enabling immediate operation.

*Figure 7.1. Python Programming  
of EyeDriveSafe*



*Figure 7.2. Product Assembly*



## **IX. System Testing and Validation**

Functional tests were conducted to verify that the USB camera successfully captured driver images, the machine learning model accurately classified alertness states, and the alert system activated upon detection of drowsiness.

*Figure 8. Drowsiness system testing*

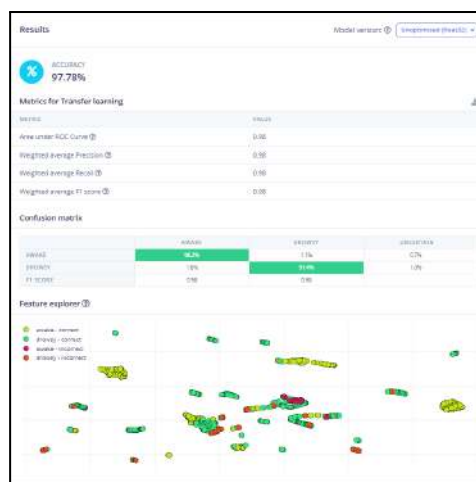


## **X. Controlled Simulation and Virtual Data Benchmarking**

In-vehicle field testing with external subjects was omitted to comply with safety regulations regarding impaired driving. System validation was instead executed through self-administered static simulations combined with virtual dataset benchmarking. Researchers

functioned as the sole test subjects within a stationary environment, intentionally executing predefined ocular behaviors to evaluate real-time hardware trigger latency, buzzer activation accuracy, and processing stability. To evaluate the transfer learning model's baseline classification performance independently of physical simulations, the algorithm was validated utilizing the automated train-test data split mechanism within the Edge Impulse platform. The model was evaluated strictly against the isolated test dataset fraction to verify precision, recall, and binary classification accuracy under standardized, reproducible parameters.

*Figure 9.1. Model Testing Performance*



*Figure 9.2 Product Simulation*



## **XI. Data Analysis or Statistical Treatment**

Data analysis and statistical treatment were conducted to determine the accuracy and reliability of the EyeDriveSafe system. Edge Impulse platform metrics, including classification accuracy, AUC-ROC, precision, recall, and F1-score, were used for model evaluation. A structured observation checklist was also utilized during validation to compare actual driver conditions with system outputs. Descriptive statistics, particularly frequency counts and percentage accuracy, were applied to assess correct detections. The consistency between observed behavior and system predictions confirmed the reliability of the results.

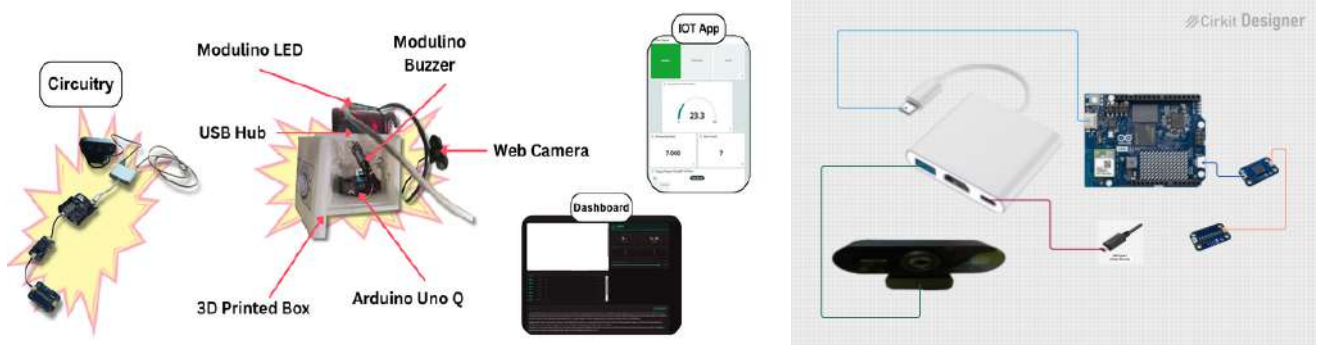
### III. RESULTS AND DISCUSSION

#### A. Proposed Product

EyeDriveSafe is a compact, dashboard-mountable drowsiness detection device housed in a 3D-printed enclosure measuring  $9.6 \times 8.2 \times 7.57$  cm externally, with internal dimensions of  $9.0 \times 7.0 \times 6.37$  cm, sufficient to securely accommodate the Arduino Uno Q and USB-C hub while minimizing component movement. A snug-fit sliding lid protects internal components during operation. A miniature car model is mounted on top of the enclosure as a design feature, with the Modulino Pixels LED strip integrated into the model's roof, providing color-coded visual alerts (green for AWAKE, yellow for WARNING, red for ALERT) that are clearly visible to the driver without requiring direct attention to the device body.

Beyond on-device detection, a local Flask web dashboard streams live detection metrics and provides an adjustable confidence threshold slider for calibrating sensitivity without retraining the model. Arduino IoT Cloud integration via MQTT enables remote monitoring of alert state, Drowsy Frame Ratio, Drowsy Episodes, and alert count, while an LLM-powered fatigue report generated through GPT-4.1-mini is delivered via the Cloud Messenger widget for post-session analysis.

*Figure 10. Final Product and Diagram of EyeDriveSafe*



The deployed deep learning model is MobileNetV2 0.35, trained on 8,930 images (7,083 training / 1,847 test) via Edge Impulse's Transfer Learning block. Key neural network settings are summarized in Table 1 below.

*Table 1. Neural Network Settings (Edge Impulse)*

| Parameter           | Value             |
|---------------------|-------------------|
| Architecture        | MobileNetV2 0.35  |
| Input resolution    | 96 x 96 greyscale |
| Input features      | 76,800            |
| Training cycles     | 400               |
| Learning rate       | 0.00003           |
| Batch size          | 32                |
| Final layer neurons | 90                |
| Dropout rate        | 0.599             |
| Data augmentation   | Enabled           |
| Training processor  | GPU               |
| Validation accuracy | 97.7%             |
| Validation loss     | 0.07              |
| Output classes      | 2 (awake, drowsy) |

## B. Performance Test Results

*Table 2. Drowsiness Detection Accuracy*

| Trial | Actual State | Condition | Awake (%) | Drowsy (%) | Classification | State Machine Output | Correct |
|-------|--------------|-----------|-----------|------------|----------------|----------------------|---------|
| 1     | Drowsy       | Daylight  | 1%        | 99%        | Drowsy         | ALERT                | ✓       |
| 2     | Awake        | Daylight  | 96%       | 4%         | Awake          | AWAKE                | ✓       |

|    |        |          |      |     |        |       |   |
|----|--------|----------|------|-----|--------|-------|---|
| 3  | Drowsy | Daylight | 12%  | 88% | Drowsy | ALERT | ✓ |
| 4  | Drowsy | Daylight | 7%   | 93% | Drowsy | ALERT | ✓ |
| 5  | Awake  | Daylight | 85%  | 15% | Awake  | AWAKE | ✓ |
| 6  | Awake  | Dark     | 100% | 0%  | Awake  | AWAKE | ✓ |
| 7  | Drowsy | Dark     | 95%  | 5%  | Awake  | AWAKE | ✗ |
| 8  | Awake  | Dark     | 93%  | 7%  | Awake  | AWAKE | ✓ |
| 9  | Drowsy | Dark     | 100% | 0%  | Awake  | AWAKE | ✗ |
| 10 | Drowsy | Dark     | 93%  | 7%  | Awake  | AWAKE | ✗ |

The drowsiness detection accuracy test evaluated EyeDriveSafe's ability to correctly classify driver states across varying conditions. Out of 10 trials, the system achieved 70% correct detection (7/10 trials), with awake state classification performing consistently well, correctly identified in all instances, with confidence scores ranging from 85% to 100%. Drowsy state detection proved more challenging, with the system correctly classifying only 3 out of 6 drowsy trials, as reduced facial feature visibility in darker conditions affected the model's ability to detect fatigue-related cues. Despite this, the system demonstrated strong confidence levels in its correct classifications, with drowsy state scores reaching as high as 99% when accurately detected. These results indicate that while EyeDriveSafe reliably identifies alert drivers, the misclassification of drowsy states under low-light conditions reveals a limitation that warrants attention. Integrating an infrared or night-vision capable camera module in future development would significantly improve the system's ability to capture facial features in dark environments, thereby enhancing overall drowsiness detection accuracy and making EyeDriveSafe more dependable across all real-world driving scenarios.

#### CRITERIA FOR ALERT RESPONSE TEST

| Rating                 | 1                                  | 2                                                         | 3                                                   | 4                                                  | 5                                                         |
|------------------------|------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------|
| Alert Activation       | The buzzer and LED do not activate | Only one of the buzzer or the LED activates               | Both activate, but with noticeable delay            | Both activate within the expected delay            | Both activate promptly and consistently across all trials |
| State Machine Accuracy | The system does not change state   | The system changes state incorrectly or at the wrong time | System changes state but occasionally skips WARNING | System escalates through all three tiers correctly | System escalates correctly and resets to AWAKE reliably   |

*Table 3. Alert Response Test*

| Trial          | Alert Activation | State Machine Accuracy |
|----------------|------------------|------------------------|
| 1              | 5                | 5                      |
| 2              | 5                | 5                      |
| 3              | 4                | 5                      |
| 4              | 5                | 5                      |
| 5              | 4                | 5                      |
| 6              | 4                | 5                      |
| 7              | 5                | 5                      |
| 8              | 4                | 5                      |
| 9              | 5                | 5                      |
| 10             | 5                | 5                      |
| <b>Average</b> | 4.7              | 5                      |

The alert response test evaluated the reliability of EyeDriveSafe’s alert mechanisms and state machine transitions across 10 trials. Alert activation received an average rating of 4.7, reflecting that both the buzzer and LED consistently activated promptly and reliably, with the majority of trials achieving the highest rating and only three trials showing activation within the expected delay rather than instantaneously. State machine accuracy achieved a



perfect average rating of 5.0 across all 10 trials, confirming that the system flawlessly escalated through all three alert tiers (AWAKE, WARNING, and ALERT) in the correct sequence and reliably reset to the AWAKE state in every instance. These results demonstrate that EyeDriveSafe’s state management logic is robust and consistent, while the minor variations in alert activation ratings reflect negligible real-world hardware response differences rather than any functional deficiency. Overall, the near-perfect scores across both criteria validate the system's capacity to deliver timely and accurate alerts during continuous driver monitoring.

*Table 4. Validation Metrics from Edge Impulse*

| Metric                     | Value |
|----------------------------|-------|
| Area under ROC Curve       | 0.98  |
| Weighted average Precision | 0.98  |
| Weighted average Recall    | 0.98  |
| Weighted average F1 Score  | 0.98  |

The validation metrics from Edge Impulse demonstrate a high-performing and statistically balanced model. An Area under the ROC Curve (AUC) of 0.98 indicates a near-perfect ability to distinguish between the two target classes. This is further supported by the weighted average Precision, Recall, and F1 Score, all of which consistently reach 0.98. These results suggest that the model maintains an excellent success rate in identifying true positives while simultaneously minimizing false alarms, ensuring that the model is both reliable and highly accurate across all performance indicators.

*Table 5. Confusion Matrix from Edge Impulse*

|              | Predicted Awake | Predicted Drowsy | Uncertain |
|--------------|-----------------|------------------|-----------|
| Actual Awake | 98.2%           | 1.1%             | 0.7%      |

|               |      |       |       |
|---------------|------|-------|-------|
| Actual Drowsy | 1.6% | 97.4% | 1.0%` |
| F1 Score      | 0.98 | 0.98  |       |

The confusion matrix reveals the specific operational accuracy of the model, which is critical for a safety-oriented application. The system correctly identifies 97.4% of "Drowsy" instances, which is the most vital metric for preventing accidents caused by fatigue. Conversely, the 98.2% accuracy for the "Awake" class, with only 1.1% misclassified as Drowsy, ensures that the user is not frequently interrupted by false positives or unnecessary alerts. With high F1 Scores of 0.98 for both categories and an overall accuracy of 97.78%, the model proves to be robust and stable, showing no significant bias toward either class.

Table 6. On-Device Performance (Edge Impulse Training and Validation)

| Metric         | Value               |
|----------------|---------------------|
| Inference time | 13 ms               |
| Peak RAM usage | 1.1MB               |
| Flash usage    | 2.0MB               |
| Compiler       | EON (RAM optimized) |

From a deployment perspective, the model is highly optimized for the Arduino’s hardware constraints. By utilizing the EON Compiler, the system achieves a remarkably fast inference time of 13 ms, enabling real-time monitoring and near-instantaneous response to changes in the user’s state. Despite its high predictive accuracy, the model maintains a light hardware footprint, requiring only 1.1MB of peak RAM and 2.0MB of Flash storage. This efficiency confirms that the model’s complex intelligence is successfully condensed to fit the functional limits of the microcontroller without sacrificing performance.

Table 7. Arduino IoT Cloud Connectivity Test

| <b>Trial</b> | <b>WiFi Connected</b> | <b>Cloud Sync Delay</b> | <b>Alert State Received</b> | <b>Drowsiness Frame Ratio</b> | <b>Drowsy Episodes Received</b> | <b>AI Report Delivered</b> |
|--------------|-----------------------|-------------------------|-----------------------------|-------------------------------|---------------------------------|----------------------------|
| 1            | Yes                   | 4 seconds               | Yes                         | Yes                           | Yes                             | Yes                        |
| 2            | Yes                   | 5 seconds               | Yes                         | Yes                           | Yes                             | Yes                        |
| 3            | Yes                   | 4 seconds               | Yes                         | Yes                           | Yes                             | Yes                        |
| 4            | Yes                   | 6 seconds               | Yes                         | Yes                           | Yes                             | Yes                        |
| 5            | Yes                   | 5 seconds               | Yes                         | Yes                           | Yes                             | Yes                        |

The Arduino IoT Cloud Connectivity Test quantified data transmission reliability and network latency. Five synchronization trials were executed: Trial 1 (4-second delay), Trial 2 (5-second delay), Trial 3 (4-second delay), Trial 4 (6-second delay), and Trial 5 (5-second delay). Each trial achieved 100% success in connecting to WiFi and transmitting alert states, drowsiness frame ratios, drowsy episodes, and AI reports. The system sustained complete data integrity across all iterations, verifying the capacity for continuous cloud synchronization. The variance in synchronization delay, spanning 4 to 6 seconds, reflects standard network latency parameters. A mean delay of 4.8 seconds, paired with zero transmission failure, validates the architecture for remote logging. Autonomous safety interventions must execute locally to bypass this 4.8-second transmission latency.

*Table 8. Large Language Model Fatigue Report Validation*

| <b>Session</b>   | <b>Duration</b> | <b>Alerts During Session</b> | <b>Report Generated</b> | <b>Contextually Accurate</b> | <b>Key Recommendation Given</b>                                   |
|------------------|-----------------|------------------------------|-------------------------|------------------------------|-------------------------------------------------------------------|
| 1 (calm driving) | 5 mins          | 0                            | Yes                     | Yes                          | Continue monitoring; take breaks every two hours on longer trips. |

|                          |         |   |     |     |                                                                                           |
|--------------------------|---------|---|-----|-----|-------------------------------------------------------------------------------------------|
| 2 (simulated drowsiness) | 10 mins | 3 | Yes | Yes | Pull over and rest before continuing; avoid driving during peak drowsiness hours.         |
| 3 (mixed states)         | 15 mins | 8 | Yes | Yes | Stop driving immediately and rest for at least 20 minutes; review regular sleep schedule. |

The Large Language Model Fatigue Report Validation test assessed the system's capacity to generate contextually accurate post-drive summaries and actionable recommendations based on varying alert frequencies. Three distinct driving sessions were evaluated: a 5-minute calm driving session with 0 alerts, a 10-minute simulated drowsiness session with 3 alerts, and a 15-minute mixed states session with 8 alerts. In all trials, the system successfully generated a report and maintained 100% contextual accuracy. The generated recommendations demonstrated a direct, proportional response to the detected hazard level. The baseline state of zero alerts yielded standard preventative advice to continue monitoring and take breaks. Moderate alert frequency (3 alerts) triggered an elevated intervention to pull over and rest. High alert frequency (8 alerts) mandated an immediate critical halt and a 20-minute minimum rest period. The results confirm the language model reliably synthesizes event frequency and session duration data into scalable, context-aware safety directives.

### C. Cost of Production

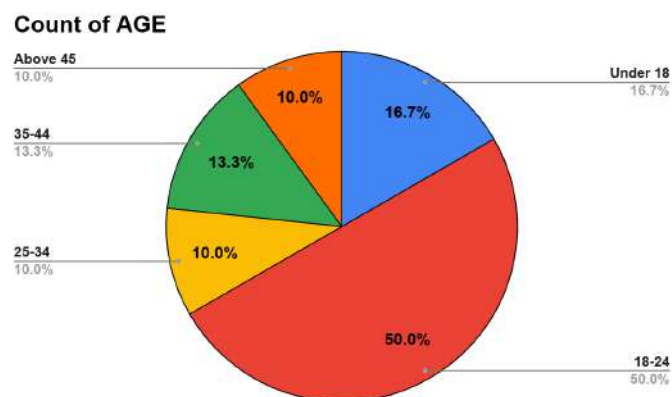
*Table 9. Cost of Production*

| No.                                  | Item/Particulars  | Quantity | Unit Price (in PHP) | Total Cost       |
|--------------------------------------|-------------------|----------|---------------------|------------------|
| 1                                    | Arduino Uno Q     | 1        | 2,574.00            | ₱2,574.00        |
| 2                                    | USB Type C Hub    | 1        | 366.00              | ₱180.00          |
| 3                                    | Web Camera        | 1        | 157.00              | ₱157.00          |
| 4                                    | Modulino Buzzer   | 1        | 382.59              | ₱382.59          |
| 5                                    | Modulino Pixel    | 1        | 580.32              | ₱580.32          |
| 6                                    | 3D Print          | 1        | 500.00              | ₱500.00          |
| 7                                    | Utility Expenses  | 1        | 0.98                | ₱0.98            |
| 8                                    | Double Sided Tape | 1        | 5.00                | ₱5.00            |
| 9                                    | Glue Stick        | 2        | 5.00                | ₱10.00           |
| 10                                   | Miniature Car     | 1        | 49.50               | ₱49.50           |
| 11                                   | Labor             |          | 86.88               | ₱86.88           |
| 12                                   | Packaging         | 1        | 100.00              | ₱100.00          |
| 13                                   | Shipping          |          | 50.00               | ₱50.00           |
| <b>Total Cost</b>                    |                   |          |                     | <b>₱4,676.27</b> |
| <b>Product Cost</b>                  |                   |          |                     | <b>₱4,676.27</b> |
| <b>Proposed Product Retail Price</b> |                   |          |                     | <b>₱6,499.99</b> |

## D. Marketability

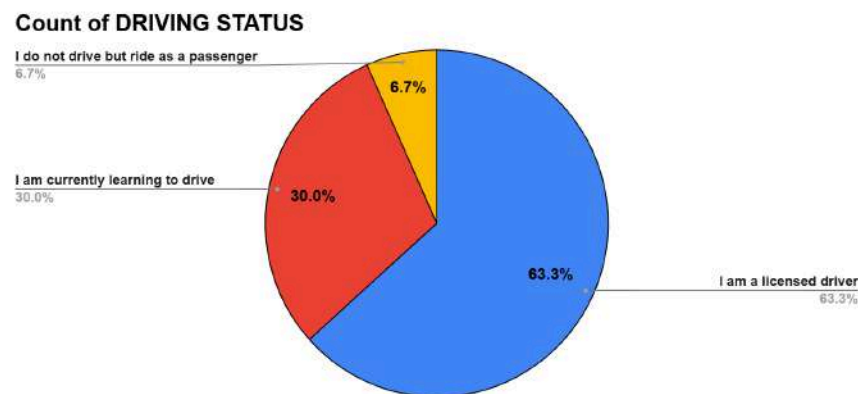
### a. Target Market Analysis

*Figure 11.1 Age Group Distribution*



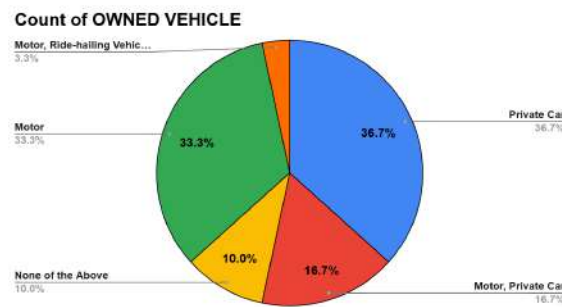
Survey results show that 50% of respondents fall within the 18–24 age group, followed by under-18 and working-age adults (25–44 and above 45). This indicates that EyeDriveSafe strongly appeals to young and emerging drivers, a group that is often exposed to fatigue due to academic demands, irregular schedules, and extended screen or driving time. As a technology-based safety device, EyeDriveSafe aligns well with this demographic’s familiarity with digital systems, increasing its likelihood of acceptance and regular use. The presence of older age groups further suggests that the product has relevance beyond youth drivers, supporting wider market applicability.

*Figure 11.2 Driving Status*



Data indicate that licensed drivers make up 63.3% of respondents, the largest segment, followed by individuals currently learning how to drive and those who ride as passengers. Licensed drivers represent the primary market for EyeDriveSafe, as they are directly responsible for vehicle operation and road safety. Meanwhile, learners and passengers represent potential future users and purchasing influencers, particularly within households. This distribution supports EyeDriveSafe’s ability to address both immediate safety needs and long-term consumer demand.

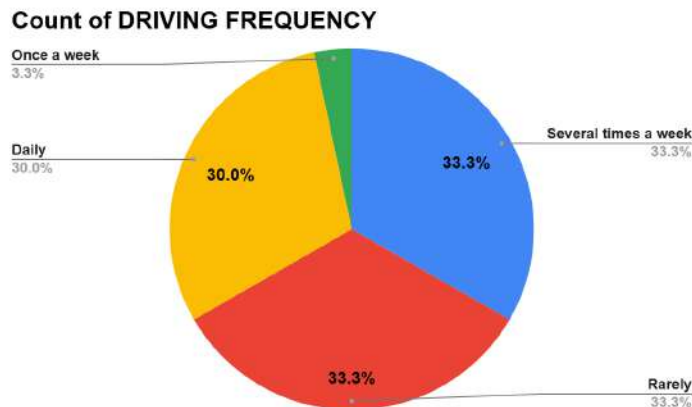
*Figure 11.3 Owned Vehicle Type*



70% respondents reported owning either a private car or a motorcycle, with 20% owning both. This confirms that EyeDriveSafe can be marketed as a versatile safety device applicable to multiple vehicle types. The inclusion of motorcycle users strengthens the product's uniqueness, as many existing drowsiness detection systems focus mainly on cars. EyeDriveSafe's portable and non-intrusive design makes it suitable for diverse vehicles, including older models that lack built-in safety technologies.

#### **b. Product Potential**

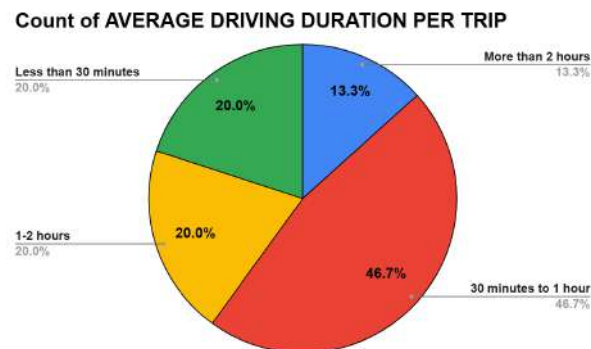
*Figure 12.1 Driving Frequency*



Survey data show that 96.6% respondents most commonly drive rarely, several times a week, or daily. The presence of frequent and daily drivers indicates continuous exposure to road conditions where fatigue may occur. This suggests strong product potential for EyeDriveSafe, as drivers who spend more time on the road are more likely to benefit from a

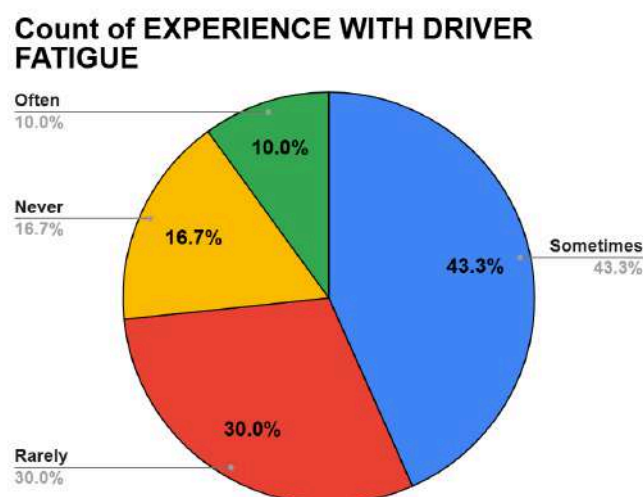
system that can detect early signs of drowsiness and provide timely alerts to prevent accidents.

*Figure 12.2 Average Driving Duration per Trip*



80% respondents reported driving between 30 minutes to more than 2 hours per trip. Longer driving durations are closely associated with increased fatigue and reduced alertness, especially during monotonous or long-distance travel. This reinforces the product potential of EyeDriveSafe, as its continuous eye-monitoring capability directly addresses risks related to extended driving periods.

*Figure 12.3 Experience with Driver Fatigue*



83.3% of respondents reported experiencing driver fatigue sometimes or often, while fewer reported rarely or never experiencing it. This indicates that drowsiness is a common driving concern among potential users. The frequent occurrence of fatigue supports the need



for EyeDriveSafe as a preventive safety device designed to alert drivers before fatigue leads to dangerous situations.

**c. Product Performance**

*Table 10. Marketability Ratings*

| <b>Marketability</b>                                                               | <b>Average Ratings</b> |
|------------------------------------------------------------------------------------|------------------------|
| EyeDriveSafe effectively detects driver drowsiness in real-time.                   | 4.72                   |
| The alert system (buzzer) responds quickly when drowsiness is detected.            | 4.68                   |
| The device is easy to use and does not distract the driver while driving.          | 4.65                   |
| EyeDriveSafe performs reliably under different lighting conditions.                | 3.93                   |
| The overall performance of EyeDriveSafe increases my confidence in driving safety. | 4.74                   |

The results presented in Table 10 indicate that EyeDriveSafe demonstrates strong product performance based on respondents' evaluations. Ratings ranged from 3.93 to 4.74, with an overall average of 4.54, reflecting a very positive perception of the system's functionality and reliability. The highest-rated statement, "The overall performance of EyeDriveSafe increases my confidence in driving safety" (4.74), indicates strong user trust in the device as an effective road safety solution. Core detection and alert functions received similarly strong ratings: real-time drowsiness detection scored 4.72, and alert responsiveness scored 4.68, confirming that the system's primary features meet user expectations.

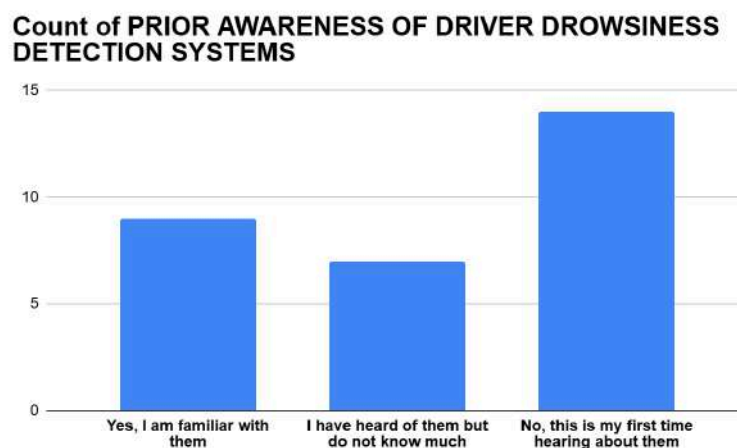
The lowest-rated indicator, "EyeDriveSafe performs reliably under different lighting conditions" (3.93), reflects a recognized limitation also documented in the drowsiness

detection accuracy (Table 2), where nighttime conditions produced the lowest detection accuracy. While 3.93 still represents a positive perception on a 5-point scale, this result highlights that lighting robustness is the area where user satisfaction diverges most from ideal performance. This is consistent with findings from the drowsiness accuracy evaluation and reinforces the recommendation for future infrared camera integration or expanded low-light training data.

Overall, the consistently high ratings across four of the five indicators confirm that EyeDriveSafe performs effectively in the eyes of its prospective users, reinforcing its marketability as a reliable and user-friendly driver drowsiness detection device.

#### d. Advertising and Marketability

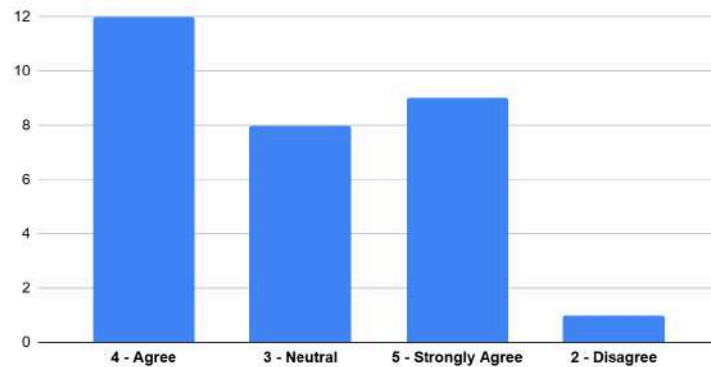
*Figure 13.1 Prior Awareness of Drowsiness Detection Systems*



Survey results show that 21 out of 30 respondents had little to no prior awareness of driver drowsiness detection systems. This suggests that EyeDriveSafe enters a market with limited exposure to similar technologies. As a result, advertising strategies should focus on educating drivers about the dangers of drowsy driving and the benefits of using EyeDriveSafe as a safety solution.

*Figure 13.2 Likelihood to Buy EyeDriveSafe*

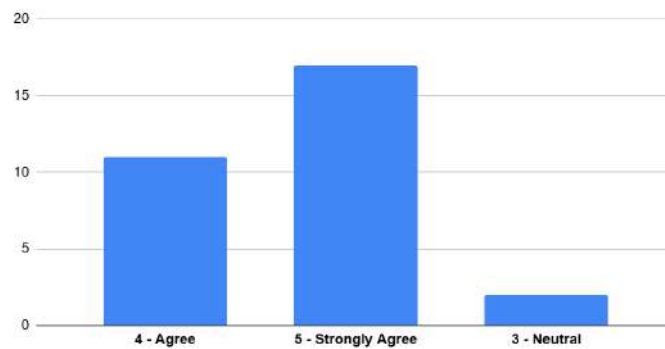
**If EyeDriveSafe were available today, how likely would you be to buy it?**



21 out of 30 respondents selected Agree or Strongly Agree when asked about their likelihood of purchasing EyeDriveSafe if it were available. This indicates strong consumer acceptance and suggests that increased awareness through advertising could effectively convert interest into actual purchases.

*Figure 13.3 Importance of Drowsiness Detection for Driving Safety*

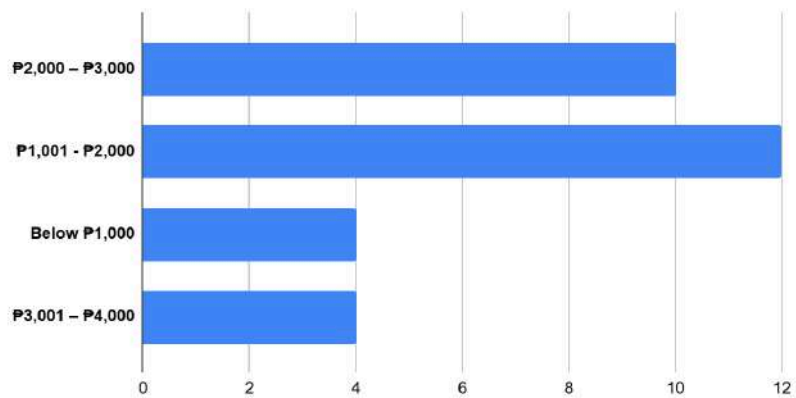
**How important is a drowsiness detection system for your driving safety?**



Survey data show that 29 out of 30 respondents rated drowsiness detection as important or very important for driving safety. This positions EyeDriveSafe as a necessity-based safety product, making it suitable for advertising that emphasizes accident prevention and driver protection rather than optional convenience features.

*Figure 13.4 Acceptable Price Range*

**Count of What price range would you consider reasonable for this product?**



The most preferred price ranges were **P1,001–P2,000** and **P2,000–P3,000**, indicating that consumers value affordability alongside functionality. Advertising EyeDriveSafe within this price range would improve accessibility and increase its appeal to a wider market.

#### IV. CONCLUSIONS AND RECOMMENDATIONS

EyeDriveSafe underwent systematic testing and evaluation, confirming that an embedded deep learning system for driver drowsiness detection is achievable within the hardware and cost constraints of the Philippine transportation context

The MobileNetV2 0.35 transfer learning model, trained on 8,930 images through Edge Impulse, achieved a validation accuracy of 97.7%, a validation loss of 0.07, and an AUC-ROC of 0.98, with weighted precision, recall, and F1-score all at 0.98. Confusion matrix analysis confirmed that 97.4% of drowsy instances and 98.2% of awake instances were correctly classified on the validation set, yielding an overall accuracy of 97.78%. On-device deployment achieved a 13-millisecond inference time at 1.1 MB of peak RAM and 2.0 MB of Flash storage, validating its efficiency for real-time embedded operation.

Rather than relying on facial landmark extraction, the system uses whole-frame binary classification to derive the Drowsy Frame Ratio (DFR) and Drowsy Episode Count (DEC), which feed a three-tier alert state machine (AWAKE-WARNING-ALERT) with time-delay false alarm prevention. Alert response testing confirmed near-perfect reliability, with state machine accuracy averaging 5.0 and alert activation averaging 4.7 out of 5 across all 10 trials.

Controlled simulation testing yielded an overall classification accuracy of 70% across 10 trials. While awake-state detection was consistent across all conditions, all three drowsy-state trials under dark conditions resulted in misclassification, representing a critical functional gap that must be addressed in future development. This limitation was reflected in consumer survey data, where lighting reliability received the lowest product performance rating of 3.93 out of 5. The proposed retail price of 6,499 Philippine pesos also exceeds

respondents' preferred range of 1,001 to 3,000 Philippine pesos, presenting a commercialization challenge requiring hardware cost reduction. Despite these limitations, EyeDriveSafe advances accessible and affordable driver safety technology for the Philippine context, aligned with SDG 3 and SDG 11.

Based on the findings from the quantitative performance evaluations of EyeDriveSafe, the researchers suggest several recommendations to improve the accuracy, reliability, and real-world applicability of the system:

1. **Model Enhancement and Dataset Expansion.** Future researchers should aim to double the current dataset size, expanding from 8,930 images to 15,000+ images. The images should focus on exploring and improving a wider range of facial behavioral patterns while also considering a wider range of conditions and environments. This expansion should also prioritize greater demographic diversity, especially in the Philippine-context, including various age groups, facial features, and skin tones, to ensure the deep learning model generalizes effectively across all populations and driving conditions.
2. **Extended Real-World Field Testing.** It is recommended that future studies conduct long-term field testing in real driving conditions, including highways, night driving, and long-distance trips. This would allow for a more comprehensive evaluation of the system's performance, durability, and user acceptance over extended periods.
3. **Hardware Optimization and Cost Reduction.** Future development may focus on optimizing hardware components to reduce production costs while maintaining performance.

4. **Adaptive State Machine Delays.** The current state machine uses fixed time delays (1.5 seconds for WARNING, 3.0 seconds for ALERT). Future versions should dynamically shorten these delays as the Drowsy Frame Ratio increases over the session, providing progressively more aggressive intervention as cumulative fatigue builds.
5. **Infrared Camera Integration for Low-Light Operation.** The controlled simulation test (Table 2) revealed that all three drowsy-state trials under dark conditions resulted in misclassification, and respondents rated lighting reliability lowest at 3.93 out of 5. Future iterations should integrate a near-infrared (NIR) camera module with dedicated IR illumination to ensure consistent facial feature visibility regardless of ambient lighting. NIR imaging operates outside the visible spectrum and does not distract the driver, directly addressing the critical detection gap identified in this study without requiring changes to the trained model architecture.
6. **Automatic Threshold Calibration Based on Environmental Conditions.** The current system provides a manually adjustable confidence threshold (default  $T = 0.50$ ) via the local web dashboard. Future development should implement automatic threshold calibration that adjusts  $T$  based on detected ambient conditions such as lighting intensity or image quality metrics. This would reduce reliance on operator intervention and maintain optimal detection sensitivity across transitioning environments, such as entering tunnels or driving from daylight into nighttime, where manual adjustment is impractical while operating a vehicle.

7. **Hybrid Landmark-Classification Architecture.** Future iterations should integrate a facial landmark detection pipeline (such as MediaPipe Face Mesh or dlib) alongside the existing transfer learning-based image classifier to form a hybrid detection system. This would combine the whole-frame classification approach used in this study with the geometric feature extraction techniques demonstrated by Mehta et al. (2019) and Essahraoui et al. (2025), enabling computation of true PERCLOS, Mouth Opening Ratio, and head pose angle. As noted by Albadawi et al. (2022), multi-indicator fusion reduces false positive rates compared to single-indicator systems, which would address the accuracy reduction observed under low-light conditions in this study.
8. **Architecture Upgrade for Enhanced AI Inference.** Future iterations of the system may benefit from migration to a platform equipped with a dedicated Neural Processing Unit (NPU), such as those found in higher-tier Qualcomm Snapdragon processors, to support more complex drowsiness detection models and achieve higher inference throughput without proportional increases in power consumption.
9. **IoT Data Buffering for Connectivity Gaps.** The current Arduino IoT Cloud integration discards session data when WiFi connectivity is unavailable (as occurs during mobile hotspot coverage gaps in transit). Future development should implement local metric buffering with batch upload upon reconnection to ensure complete session records for post-trip analysis.
10. **LLM report validation and accuracy assessment.** The AI-generated fatigue reports via GPT-4.1-mini have not been validated against expert human assessment. Future



studies should compare LLM-generated reports with sleep specialist evaluations to establish report reliability.

**11. Native Mobile Application for Real-Time Monitoring.** The current remote monitoring capability relies on the Arduino IoT Cloud web interface and Messenger widget, which require browser access and are constrained by the platform's layout and notification system. Future development should pursue a dedicated native mobile application that consolidates real-time alert state monitoring, historical session data, and LLM-generated fatigue reports into a single interface. A native application would enable background push notifications for critical alert events, offline access to past session reports, and a more responsive user experience compared to browser-based dashboards, improving the practical adoption of the system's IoT capabilities.

**12. Fleet Operator Accountability.** The IoT remote monitoring capability of EyeDriveSafe, which enables family members or fleet operators to track driver fatigue states and receive AI-generated session reports, provides a framework for fleet-level fatigue management. Transport companies and logistics operators should explore integrating drowsiness monitoring data into driver scheduling and compliance systems, using accumulated DFR and alert history to identify high-risk drivers or routes and adjust shift assignments accordingly.

**13. Social Awareness and Policy Integration.** Future initiatives should focus on strengthening public awareness and policy integration to maximize the societal impact of EyeDriveSafe and similar fatigue-detection technologies. Researchers and stakeholders are encouraged to collaborate with local government units, transportation

agencies, and road safety organizations to promote driver fatigue awareness campaigns, especially in high-risk sectors such as public transportation, delivery services, and long-haul logistics. Policymakers may also explore the development of guidelines or incentives that encourage the adoption of fatigue monitoring systems in commercial fleets, such as tax benefits, safety certifications, or inclusion in road safety compliance standards. Embedding driver drowsiness monitoring within broader road safety policies can help normalize preventive safety technologies, foster a responsible driving culture, and contribute to systemic reductions in fatigue-related road incidents.

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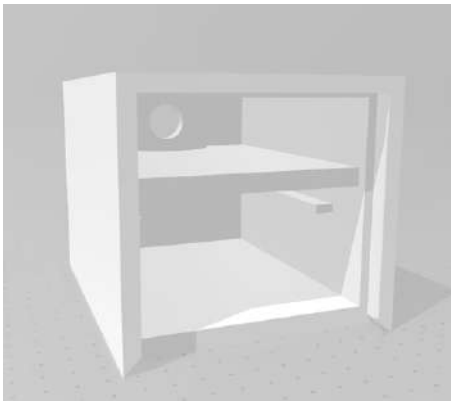
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APPENDICES

A. DOCUMENTATION

Picture 1. Initial Draft of EyeDriveSafe’s chassis



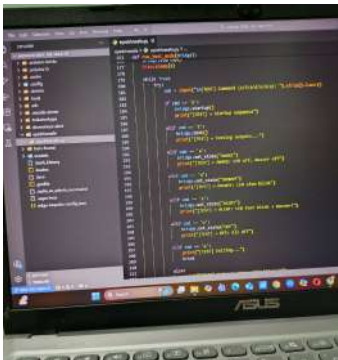
Picture 2. Invoice of Arduino Uno Q

| Order Source: |         | Order Date: |         | Invoice Date:                                                                                                    |  | Ship Date:   |  | Document Date:   |  |
|---------------|---------|-------------|---------|------------------------------------------------------------------------------------------------------------------|--|--------------|--|------------------|--|
| INTERNET      |         | 24-NOV-2025 |         | 25-NOV-2025                                                                                                      |  | 25-NOV-2025  |  | 25-NOV-2025/AUTO |  |
| Line Item     | Ordered | Cancelled   | Shipped | Item Number/Description                                                                                          |  |              |  |                  |  |
| 1             | 1       | 0           | 1       | PART: 1050-ABX00162-ND<br>MFR : Arduino / ARX00162<br>COD : TAIWAN, PROVINCE OF CHINA<br>Net Weight: 0.072291 kg |  |              |  |                  |  |
|               |         |             |         | DESC: ARDUINO                                                                                                    |  | ECON: 569920 |  | 25               |  |

Picture 3. Testing of the EyeDriveSafe



Picture 4. Portion of the EYEDRIVE code during development



Picture 5. Testing of the EyeDriveSafe



Picture 5. Google Forms for distanced target customers

**EYEDRIVESAFE: A PORTABLE IoT-ENABLED DRIVER DROWSINESS DETECTION AND MONITORING SYSTEM USING TRANSFER LEARNING-BASED FACIAL BEHAVIORAL ANALYSIS**

(Draft only)

Our research group from Irvine is currently conducting a study titled "EYEDRIVESAFE: A Portable Driver Drowsiness Detection System Using Deep Neural Network Based Monitoring". In accordance with the data privacy policy of UC Irvine (UCIR) IRB No. 2017-05, we would appreciate that all personal information collected will be handled with utmost confidentiality, security, safety, and used solely for academic and research purposes.

The purpose of this study is to develop and evaluate EYEDRIVESAFE, a deep neural network-based driver drowsiness detection system that utilizes real-time non-invasive monitoring through camera-based facial analysis to help prevent road accidents caused by driver fatigue. Specifically, this research aims to monitor the system's detection accuracy, response time, adaptability across varying lighting conditions, and potential consumer acceptance in real-world driving scenarios. To achieve this, we have prepared a survey questionnaire to gather the necessary data from our target respondents.

Participation in this study is entirely voluntary. You may choose not to answer any question or withdraw at any time without penalty. All collected data will only be accessed by the researchers and their advisors and will be permanently deleted after the completion and presentation of the study. Your participation is highly valuable and will greatly contribute to the success of this research. Thank you very much for your time and cooperation.

anyone@uci.edu | uci.edu/eyedrive

Not shared

Required to view

Optional required question

By proceeding with this survey, you confirm that you have read and understood the information provided above, including the purpose of the study and the provisions of the Data Privacy Act of 2012 (RA 10173). You also confirm that you voluntarily agree to participate in this research.

☒ Yes, I have read and understood the information above, and I give my consent to participate in this study.

☐ No, I do not give my consent to participate in this study.

Picture 6. Connection of Components



Picture 7. Mounting the Miniature Car



Picture 8. Designing EyeDriveSafe



Picture 9. Putting sticker paper on EyeDriveSafe



Picture 10. Final Product



Picture 11. Consent letters signed by Respondents



Republic of the Philippines  
**Department of Education**  
NATIONAL CAPITAL REGION  
SCHOOLS DIVISION OF TAGUIG CITY AND PATEROS  
**TAGUIG SCIENCE HIGH SCHOOL**

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To our participating volunteers,

Greetings!

The researchers from Grade 12 - Hawking of Taguig Science High School sincerely seek your consent to participate in our ongoing study titled **"EYEDRIVESAFE: A PORTABLE IoT-ENABLED DRIVER DROWSINESS DETECTION AND MONITORING SYSTEM USING TRANSFER LEARNING-BASED FACIAL BEHAVIORAL ANALYSIS"**, in partial fulfillment of the requirements for the subject Inquiries, Investigations, and Immersion.

The researchers request your permission to take your photograph and upload it to the EyeDriveSafe model for training and validation purposes. The researchers assure you that all photographs will remain confidential and private, and will be used solely for this study. Your photographs will greatly contribute to the accuracy of the EyeDriveSafe of our research.

The researchers appreciate your support and participation to this ongoing study. Thank you very much!

Respectfully yours,  
The researchers from  
Grade 12 - Hawking of Taguig Science High School

Signature of volunteers



---





M.L. Quezon Ave., Brgy. San Miguel, Taguig City  
(02) 8244-5346  
tagscihs2004@deped.gov.ph

Picture 12. First Page of the Survey Questionnaire

**EYEDRIVESAFE: A PORTABLE IoT-ENABLED DRIVER DROWSINESS DETECTION AND MONITORING  
SYSTEM USING TRANSFER LEARNING-BASED FACIAL BEHAVIORAL ANALYSIS**

**Survey Form**

Thank you for taking the time to participate in this survey. Your feedback is valuable in assessing and improving EYEDRIVESAFE, a portable driver drowsiness detection system that utilizes deep learning-based monitoring and eye-closure analysis. Please answer all questions honestly. Kindly shade the circle and place check marks in the provided tables below.

| GENERAL INFORMATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    | TARGET MARKET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |              |                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|-----------------------|
| Name: _____<br>Age:<br><input type="radio"/> Under 18<br><input type="radio"/> 18 - 24<br><input type="radio"/> 25 - 34<br><input type="radio"/> 35 - 44<br><input type="radio"/> Above 45<br>Gender:<br><input type="radio"/> Male<br><input type="radio"/> Female<br><input type="radio"/> Prefer not to say                                                                                                                                                                                                                                                 |                    | 1. Owned Vehicle (Select all that apply):<br><input type="radio"/> Motor<br><input type="radio"/> Private Car<br><input type="radio"/> Public Utility Vehicle (e.g., jeepney, bus, taxi)<br><input type="radio"/> Ride-hailing Vehicle (e.g., Grab, Angkas)<br>2. Driving Status<br><input type="radio"/> I am a licensed driver<br><input type="radio"/> I am currently learning to drive<br><input type="radio"/> I do not drive but ride as a passenger<br>3. How often do you drive long distances per month?<br><input type="radio"/> 0-2<br><input type="radio"/> 3-5<br><input type="radio"/> 6-8<br><input type="radio"/> 9-10<br><input type="radio"/> More than 10 |             |              |                       |
| PRODUCT POTENTIAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |              |                       |
| 1. Driving Frequency<br><input type="radio"/> Daily<br><input type="radio"/> Several times a week<br><input type="radio"/> Once a week<br><input type="radio"/> Rarely<br>2. Average Driving Duration per Trip<br><input type="radio"/> I am a licensed driver<br><input type="radio"/> I am currently learning to drive<br><input type="radio"/> I do not drive but ride as a passenger<br>3. Experience with Driver Fatigue<br><input type="radio"/> Often<br><input type="radio"/> Sometimes<br><input type="radio"/> Rarely<br><input type="radio"/> Never |                    | 4. What do you like most about the EyeDriveSafe concept?<br>_____<br>_____<br>5. What concerns or improvements would you suggest for EyeDriveSafe?<br>_____<br>_____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |              |                       |
| ADVERTISING AND MARKETABILITY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |              |                       |
| 1. Prior Awareness of Driver Drowsiness Detection Systems<br><input type="radio"/> Yes, I am familiar with them<br><input type="radio"/> I have heard of them but do not know much<br><input type="radio"/> No, this is my first time hearing about them<br>2. Average Driving Duration per Trip<br><input type="radio"/> I am a licensed driver<br><input type="radio"/> I am currently learning to drive<br><input type="radio"/> I do not drive but ride as a passenger                                                                                     |                    | 3. Experience with Driver Fatigue<br><input type="radio"/> Often<br><input type="radio"/> Sometimes<br><input type="radio"/> Rarely<br><input type="radio"/> Never<br>4. What price range would you consider reasonable for this product?<br><input type="radio"/> Below ₱1,000<br><input type="radio"/> ₱1,001 - ₱2,000<br><input type="radio"/> ₱2,000 - ₱3,000<br><input type="radio"/> ₱3,001 - ₱4,000<br><input type="radio"/> Above ₱5,000                                                                                                                                                                                                                             |             |              |                       |
| STATEMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5 - Strongly Agree | 4 - Agree                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3 - Neutral | 2 - Disagree | 1 - Strongly Disagree |
| Overall, what is your impression of the EyeDriveSafe concept?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |              |                       |
| How interested would you be in using a device like EyeDriveSafe in your vehicle?                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |              |                       |



Picture 13. Second Page of the Survey Questionnaire

|                                                                          |  |  |  |  |  |
|--------------------------------------------------------------------------|--|--|--|--|--|
| If EyeDriveSafe were available today, how likely would you be to buy it? |  |  |  |  |  |
| How important is a drowsiness detection system for your driving safety?  |  |  |  |  |  |

---

SIGNATURE OVER PRINTED NAME

Picture 14. Survey Questionnaire for demo participants

**EYEDRIVESAFE: A PORTABLE IoT-ENABLED DRIVER DROWSINESS DETECTION AND MONITORING  
SYSTEM USING TRANSFER LEARNING-BASED FACIAL BEHAVIORAL ANALYSIS**

Survey Form

Thank you for taking the time to participate in this survey. Your feedback is valuable in assessing and improving EYEDRIVESAFE, a portable driver drowsiness detection system that utilizes deep learning-based monitoring and eye-closure analysis. Please answer all questions honestly. Kindly shade the circle and place check marks in the provided tables below.

| PRODUCT PERFORMANCE                                                                |                    |           |             |              |                       |
|------------------------------------------------------------------------------------|--------------------|-----------|-------------|--------------|-----------------------|
| STATEMENTS                                                                         | 5 - Strongly Agree | 4 - Agree | 3 - Neutral | 2 - Disagree | 1 - Strongly Disagree |
| EYEDRIVESAFE effectively detects driver drowsiness in real-time.                   |                    |           |             |              |                       |
| The alert system (buzzer) responds quickly when drowsiness is detected.            |                    |           |             |              |                       |
| The device is easy to use and does not distract the driver while driving.          |                    |           |             |              |                       |
| EYEDRIVESAFE performs reliably under different lighting conditions.                |                    |           |             |              |                       |
| The overall performance of EYEDRIVESAFE increases my confidence in driving safety. |                    |           |             |              |                       |

---

SIGNATURE OVER PRINTED NAME

## B. RESEARCHER'S PROFILE INFORMATION



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